



Rangamati Science and Technology University

Department of Computer Science and Engineering

4th Year 2nd Semester B.Sc. (Engg.) Final Exam-2023

Course Code: CSE 4202; Session: 2019-2020

Course Title: Engineering Management

Time: 3 Hours

Marks: 60

- NB: 1. Answer any **FOUR (4)** questions out of **SIX (6)** questions.
2. Figures in the right margin indicate marks($15 \times 4 = 60$).
3. All parts of a question must be answered serially.

1. a) What is management? Briefly discuss the core functions of management. 1+4=5
b) Describe Katz's three essential managerial skills and how the importance of these skills changes depending on managerial level. 2+2=4
c) Briefly discuss the scope of engineering management in light of the "Universality of Management" concept. 6
2. a) What is planning? Why managers do plan? 2+2=4
b) Explain Mintzberg's managerial roles and discuss how these roles help managers perform effectively in an organization. 5
c) Why planning is treated as the most important steps of management? Discuss the answer from an engineer's point of view. 6
3. a) Leaders are born, they are not made"-Discuss briefly. 5
b) Describe each of the five forms of departmentalization with advantages and disadvantages. 6
c) What do you mean by chain of command, authority, responsibility and unity of command? 4
4. a) What is MBO? Briefly explain the benefits and weakness of MBO. 5
b) Discuss the dual nature of a leader's behavior by explaining the managerial grid. 6
c) Differentiate between transactional and transformational leaders. 4
5. a) Write short notes on Path-Goal Theory 4
b) Describe the five levels in Maslow's hierarchy and how it can be used in motivation efforts. 6
c) Broadly discuss the "Two Factor Theory" of motivation. 5
6. a) What do you mean by controlling? State the reasons why control is important 2+3=5
b) Explain the planning-controlling link. 6
c) Differentiate between feed forward and feedback control as controlling tools. 4





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4th Year 2nd Semester B.Sc. (Engg.) Final Exam -2023

Course Code: CSE 4203; Session: 2019-2020

Course Title: Digital Systems Design

Time: 3 Hours

Marks: 60

- NB: 1. Answer any **FOUR (4)** questions out of **SIX (6)** questions.
2. Figures in the right margin indicate marks (15*4=60).
3. All parts of a question must be answered serially.

1. (a) Define Register Transfer Language (RTL). Explain conditional and parallel microoperations with suitable sketches and examples. **05**
- (b) The register values are: **04**
R1 = 1010 1101, R2 = 0110 1001
Perform the following microoperations with all intermediate steps and final register values:
i) **R3 ← R1 + R2**
ii) **R1 ← shr R1**
iii) **If R2(7)=1 then R4 ← R1**
iv) **R2 ← shl R2**
- (c) Explain how memory reading and memory write operation are represented using RTL. **04**
- (d) Given the control condition **T=1**, execute the following microoperation: **02**
T: R2 ← R1, R1 ← R2
If initial **R1 = 1100** and **R2 = 001**, what are the final contents of **R1** and **R2** after the operation?
2. (a) What are the two main ways a common bus system can be constructed? Briefly describe their fundamental operational mechanism with circuit diagram. **10**
- (b) Suppose a computer uses a bus controlled by select lines **S₂S₁S₀**. **05**
AC = 1100 1111 0101 1010
DR = 0000 1111 0000 1111
When **S₂S₁S₀ = 100**, **AC** is placed on the bus.
Perform:
i) **Bus ← AC**
ii) **DR ← Bus**
iii) **AC ← AC + DR**
Show the binary addition and final **AC**.

3. (a) Write Verilog code for a binary adder constructed using full-adder (FA) circuits to perform the arithmetic sum of two binary numbers. **05**
- (b) Suppose a 4-bit register has the binary value **0110**. It will become **0111** after it is incremented. Design a 4-bit combinational circuit for the increment microoperation. **05**
- (c) How can addition and subtraction operations be combined into a single circuit? Explain with a circuit diagram. **05**

- 4 (a) Design circuit of arithmetic microoperations for performing following operation: 07

$$R3 \leftarrow R1 + \overline{R2} + 1$$
- (b) What is Logic Micro-operations? Explain what logic micro-operations are. Then, design a single-stage combinational circuit that can perform the four basic logic micro-operations — **AND**, **OR**, **XOR**, and **NOT** — on two binary operands, A and B. Explain how all other micro-operations can be derived from these four basic operations. 08
5. (a) Compare Hardwired Control and Microprogrammed Control. 03
- (b) List the inputs and outputs of the hardwired Control Logic Gates in Mano's computer. 02
- (c) If the computer has 16 distinct instructions, propose a mapping process to jump to the correct microprogram routine. 05
- (d) In microprogrammed control, describe the function of the **Control Memory Address Register (CAR)** and the **Control Data Register (CDR)**. 05
6. (a) Which memory technology generally uses the least power? What are the main drawbacks of dynamic RAM compared with static RAM? 04
- (b) A certain memory has a capacity of **4K X 8**. 03
- (a) How many data input and data output lines does it have?
- (b) How many address lines does it have?
- (c) What is its capacity in bytes?
- (c) What is the function of the R/W input? Outline the steps that take place when the CPU reads from memory. 04
- (d) Explain the expanding capacity of computer memory. 04



Rangamati Science and Technology University

Department of Computer Science and Engineering

4th Year 2nd Semester B.Sc.(Engg.) Final Examination, 2023

Session: 2019-2020

Course Title: Information Security

Course Code: CSE-4205

Time: 3 Hours

Full Marks: 60

[Answer any Four of the following questions. Figures in the right-hand margin indicate full marks.]

1.
 - a) Define the snooping and spoofing. Discuss the ethical concepts in Information Security. 04
 - b) Explain the NSTISSC security model and the top down approach to security Implementation. 05
 - c) Suppose that you have selected a system in which $p = 131$, $q = 151$, and $e = 11143$. Verify that the conditions of the RSA system are satisfied. Compute $ke(\cdot)$ and $kd(\cdot)$. Which positive integer less than $p \cdot q = 19781$ was the secret message if the encoded message received is 141? (You will find the results of an earlier exercise useful.) 06

2.
 - a) What kind of security **Threats & Attacks** that can affect mobile devices has face due to base of security? Explain it. 04
 - b) Find the output of the initial permutation box when the 64-bit plaintext input block is given in hexadecimal as:

$$0x0002\ 0000\ 0000\ 0001$$

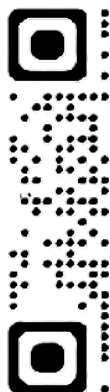
03
 - c) Suppose that $q = 23$ and $a = 7$. Find the value of K for both Alice and Bob which is the symmetric shared key in the Diffie-Hellman protocol. 04
 - d) Explain the types of transformations that which is used by (AES) **Advanced Encryption Standard** To provide security. 04

3.
 - a) Explain the Approach process of Fixed password-based authentication for authentication. 03
 - b) Distinguish between: 2*3=06
 - a. Confusion and Diffusion
 - b. Active attacks & Passive attacks
 - c. Phreaking & Phishing
 - c) What is SSL Handshake protocol? Derive the work Scheme of SSL step by step with proper Explanations? 03
 - d) Write about key generation, encryption and decryption in ElGamal Cryptosystem. 03

4.
 - a) Write the structure scheme of SHA and Whirlpool in designing a hash function. 04



- b) Describe the process steps of Merkle-Damgard scheme as the basis for iterated hash functions. 04
- c) Derive the creation of words for the AES-128 version key-expansion process with proper example. 04
- d) In Rabin Cryptosystem, Suppose $p = 23$ & $q = 7$, Calculate the value of n, P, c to find the answer. 03
5. a) Write the authentication dialogue exchanged between a user and authentication server in Kerberos. 03
- b) Describe the modes of operation that are used by intrusion detection and prevention systems. 03
- c) Distinguish between : 06
- I. Feistel ciphers Vs. Non-Feistel ciphers
 - II. RSA Vs. Rabin Cryptosystems
 - III. Discrete logarithm attack & Man-in-the-middle attack
- d) Derive the process of creating a session key established between Alice and Bob for communication using Needham-Schroeder protocol. 03
6. a) Write down the steps of Security Requirements. 03
- b) What is the difference between physical cluster and virtual cluster? 04
- c) Discuss the cloud architecture with suitable block diagram. 04
- d) Write what you know about Digital Certificate in your opinion. Describe the Classification of Digital Certificate. 04





Rangamati Science and Technology University

Department of Computer Science and Engineering

4th Year 2nd Semester B.Sc. (Engg.) Final Exam-2023

Course Code: CSE-5210; Session: 2019-2020

Course Title: Machine Learning

Time: 3 Hours

Marks: 60

- NB:
1. Answer any **FOUR (4)** questions out of **SIX (6)** questions.
 2. Figures in the right margin indicate marks ($15 \times 4 = 60$).
 3. All parts of a question must be answered serially.

1. (a) Explain the characteristics of an artificial neural networks (ANN). 3
 (b) Define Machine Learning. What's the difference between unsupervised learning and supervised learning? 4
 (c) How to handle missing and duplicate values in machine learning model training? 4
 (d) Shortly brief the Adaptive Neuro-Fuzzy Inference System (ANFIS) with diagram. 4
2. (a) Define employed bees and Scout bees. 3
 (b) Considering following path graph and graph of pheromone using Ant Colony Optimization algorithm we get probability of selecting a path taking $\alpha \neq \beta \neq 1$. 4

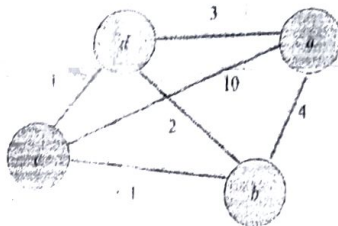


Fig a: Length of path

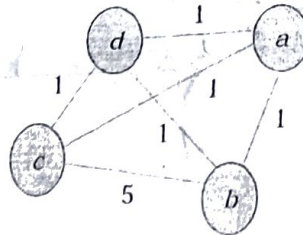


Fig b: Concentration of pheromone

- (c) State the application of swarm intelligence. Draw the flow chart of particle swarm optimization (PSO) algorithm. 4
 (d) Explain the velocity and position update of i th particle at Particle swarm optimization with appropriate equation. 4
3. (a) What are false positives and false negatives? Why are they significant? 4
 (b) Given, $U = \{p, q, r, s\}$, where A and B are the Fuzzy sets associated with U. Here $\mu_A(p) = 0.2$, $\mu_A(q) = 0.7$, $\mu_A(r) = 0.18$, $\mu_A(s) = 0.15$, $\mu_B(p) = 0.3$, $\mu_B(q) = 0.6$, $\mu_B(r) = 0.15$, $\mu_B(s) = 0.1$. 5
 Now find the Fuzzy sets $A \cup B$, $A \cap B$, A^c and B^c .
 (c) Consider an objective function, $f(x) = x^5 - 11x^4 + 27x^3 - 49x^2 + 30x$ and the initial values of five food sources are given in table below. Apply ABC (Artificial bee colony) to find minimum value of the function (complete only employed bees and onlooker bees phase). 6

Food	2.5307	0.8182	0.8993	2.1095	1.9673
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4. (a) Explain the concept of hard margin SVM. Define support vectors and discuss why they are critical in finding the optimal hyperplane. 5
- (b) Points (1, 2), (2, 3), (3, 3), (4, 5) with labels $y = [+1, -1, 1, 1]$. Compute the weight vector w and bias b step by step for a linear SVM with hard margin. 5
- (c) Given SVM decision scores [1.5, -0.8, 0.6, -1.2] for a test set, compute predicted labels using the sign function. Determine the number of correct predictions if true labels are [-1, -1, 1, 1]. 5
5. (a) Shortly describe the several techniques of selection of population for mating pool in machine learning evolutionary algorithms. 5
- (b) Write down the six steps of genetic algorithm in EC and explain the steps. 5
- (c) What is evolutionary computation? Let us consider an objective function, $f(x, y) = x^2 + y^2$, we have to find the minima of it. Show the process pictorially. 5
6. (a) Define fuzzy set and fuzzy variables with examples. 3
- (b) Why dimensionality reduction is useful in machine learning? 4
- (c) State the basic concepts of Radial Basis function (RBF) and De-fuzzification techniques. 4
- (d) Find out the object for the following figure and parameters. Consider that the object has Propeller (+1), Engine (+1) and Stern (+1), but does not have Wings (-1). 4

